

EdgeQuake capacity & workload — plain English

A full hybrid measurement: local system ingest with a local language model (Ollama), plus live question-and-answer on the shared demo tenant. Numbers are evidence, not marketing targets.

RUN ID

2026-09-29-full-hybrid

SUITE

Hybrid (System local + Provider demo)

DEMO URL

https://demo.edgequake.com

DEMO VERSION

0.28.3

TENANT

00000000-0000-0000-0000-000000000002

WORKSPACE

00000000-0000-0000-0000-000000000003

Bottom line

- On the demo tenant, fresh unique questions completed with success rate 100% and typical server time about 10779.5 ms (upper ~15600.0 ms).
- Repeated questions under concurrency 4 showed answer-cache hit rate 1.0 and typical time 2198.0 ms.
- Local mock ingest (short docs) completed at ~0.0296 docs/s at concurrency 1 (system bench, not live LLM).
- No capacity knee found up to concurrency 8 under the 25s comfort hypothesis for cold unique prompts.

1. What we tested

EdgeQuake turns documents into a searchable knowledge graph, then answers questions. Because answers depend on language models, a single “requests per second” number can mislead. This report follows SPEC-153: measure each service layer, keep *fresh* answers separate from *cached* repeats, and say clearly what was not tested.

PART	WHERE	WHAT
System cards S1–S4	Local mock backend	Admit/queue proxy, short & dense ingest, embedding estimate
Provider P1 cold	demo.edgequake.com	Unique questions, concurrency 1→8, 25s comfort target
Provider P2 warm	demo.edgequake.com	Repeated questions (answer cache)
Provider P3 stream	demo.edgequake.com	Streaming path / time-to-first-token signals

Not tested on the shared demo: uploading or re-processing PDFs (ingest L2–L6 with a live vendor). That protects the shared corpus and matches SPEC-153 safety rules. Ingest capacity below is *local mock* only.

2. Demo question experience (live)

FRESH SUCCESS 100% Unique cold prompts at C=1	FRESH TYPICAL TIME 10.8 s Server total_time p50	FRESH UPPER TIME 15.6 s About 90% of asks faster than this
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USERS AT ONCE	ASKS	SUCCESS	IN SLO	TYPICAL	UPPER	CACHE HITS
1	6	100%	100%	10.8 s	15.6 s	0%
2	6	100%	100%	7.9 s	11.3 s	0%
4	8	100%	100%	11.6 s	13.5 s	0%
8	8	100%	100%	15.4 s	18.8 s	0%

Warm (repeats): concurrency 4 · success 100% · cache hit rate 100% · typical 2.2 s · upper 3.9 s.

Streaming: samples=4 · TTFT p50=0 ms · TTFT p90=0 ms.

Capacity knee: not_reached_in_range — tested up to C=8.

3. Local ingest (system / mock model)

These numbers show how the EdgeQuake pipeline moves documents when the language model is a fast mock. They isolate software and database work from vendor LLM latency.

Admit / queue proxy (S1)

16 tiny uploads · 30.48 admits/s · admit p50 33 ms · p90 44 ms.

Short documents (S2)

CONCURRENCY	SUCCESS RATE	DOCS/S	TYPICAL SOJOURN	UPPER
1	1.0	0.0296	31.3 s	40.9 s
2	1.0	0.0375	48.0 s	56.3 s

Dense documents (S3)

CONCURRENCY	DOCS/S	TYPICAL SOJOURN
1	0.0149	67.0 s

Embedding estimate (S4)

Estimated embedding tokens (chars÷2.5): 2576 · approx 51.83 est-tokens/s over ingest wall time · oracle_ok=True. This is an **estimate**, not provider usage.

4. Layer scorecard

LAYER	STATUS	PLAIN-ENGLISH CLAIM
L0	measured	API accepted authenticated Q&A traffic for the pinned tenant
L1	measured	Local admit path handled burst uploads (queue physics detail in SPEC-090)
L2	gap	No PDF convert capacity claim
L3	measured	Chunking completed as part of local mock ingest
L4	measured	Local mock extraction completed; not live LLM capacity
L5	estimated	Embedding work reported as estimate (chars/2.5), not provider usage
L6	measured	Local merge/persist finished for completed documents
L7	measured	Demo retrieval participated in successful Q&A
L8	measured	Fresh demo answers typically seconds-scale; repeats may be cached

5. What stakeholders can say

Safe claims

- Demo Q&A succeeded for sampled unique prompts on the pinned tenant/workspace

- Cold and warm query modes are reported separately
- Local system ingest with mock LLM completed for short/dense shapes
- Embedding rates are labeled estimates

Avoid saying

- Do not merge warm cache throughput into cold LLM capacity
- Do not claim Vision/PDF token capacity
- Do not treat health llm_provider=true as a capacity probe
- Do not claim shared-demo Provider ingest capacity (not run)

6. Method appendix

- Protocol: specs/153-workload-benchmark/ (SPEC-153).
- Comfort hypothesis for demo Q&A: total server time ≤ 25 seconds; attainment $\geq 90\%$.
- Cold mode on hosted demo uses unique prompts because server cache cannot be force-disabled from the client.
- Pass/fail: report card complete and internally consistent — not “hit a marketing tokens/sec.”
- Evidence: specs/153-workload-benchmark/measurements/2026-09-29-full-hybrid/

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